- There are 4 Questions for 15 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class. You may use any result from class directly (without re-deriving the proof).
- You do not need any calculators.
- Good luck!

1. [3] Suppose $A \in \mathbb{R}^{m \times n}$ is a fat matrix (i.e. $m < n$).
   (a) Can the corresponding map $f(\underline{x}) = A\underline{x}$ be onto ? Justify.
   (b) Can the corresponding map $f(\underline{x}) = A\underline{x}$ be one-one ? Justify.

2. [3] Suppose $A \in \mathbb{R}^{m \times n}$ is an $m \times n$ matrix, and $f(\underline{x}) = A\underline{x}$ the corresponding map. For each of the statements below, answer if they are true or false with justification.
   (a) If f is one-one, then the system $A\underline{x} = \underline{b}$ has either no solution or a unique solution.
   (b) If f is onto, then the system $A\underline{x} = \underline{b}$ always has a solution (regardless of $\underline{b}$): though it is possible it may have infinitely many solutions.

3. [4½] Suppose the columns of matrix A are linearly independent, and so are the columns of the matrix B.
   (a) Suppose A and B have the same number of rows, and consider the stacked matrix $M = \begin{pmatrix} A & B \end{pmatrix}$. Are the columns of M linearly independent?
   (b) Suppose A and B have the same number of columns and consider the stacked matrix $M = \begin{pmatrix} A \\ B \end{pmatrix}$. Are the columns of M linearly independent?
   (c) Suppose A is $m \times n$ and B is $p \times q$. Consider the block matrix

$$M = \begin{pmatrix} A & 0_{m \times q} \\ 0_{p \times n} & B \end{pmatrix}$$

   Are the columns of M linearly independent ? Here $0_{a \times b}$ is a zero matrix of size $a \times b$.

4. [4½] Consider a graph with three nodes and three edges, where every node is connected to every other node. What is the adjacency matrix of this graph ? Are the columns of the adjacency matrix linearly independent ?

   Recall that a graph's adjacency matrix is a square matrix whose entry in position (i, j) is 1 if node i is connected to node j, 0 otherwise.